



# Active Learning for Multidialectal Arabic POS Tagging



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Open Source

Arabic NLP  
Tools and Datasets



## Resources

Download and demo our tools and datasets

+ Arabic Ontology

الأنطولوجيا العربية

+ Lexicographic Databases (Qabas)

حوسبة المعاجم (قبس و150 معجم)

+ Dialect Corpora (Currasat)

مدونة اللهجات العامية (كراسات)

+ SinaTools

أدوات سينا

+ Morphology Tagger (Alma)

المحلل الصرفي (ألمى)

+ Word Sense Disambiguation (Salma)

المحلل الدلالي (سلمى)

+ Named Entity Recognition (Wojood)

استخراج أسماء الاعلام (وجود)

+ Relation Extraction

استخراج العلاقات

+ Social Computing (Fada)

الإنسانيات الحاسوبية والتواصل الاجتماعي (فضا)

+ Synonyms

استخراج المترادفات

+ Chatbots and intent detection (AraBanking77)

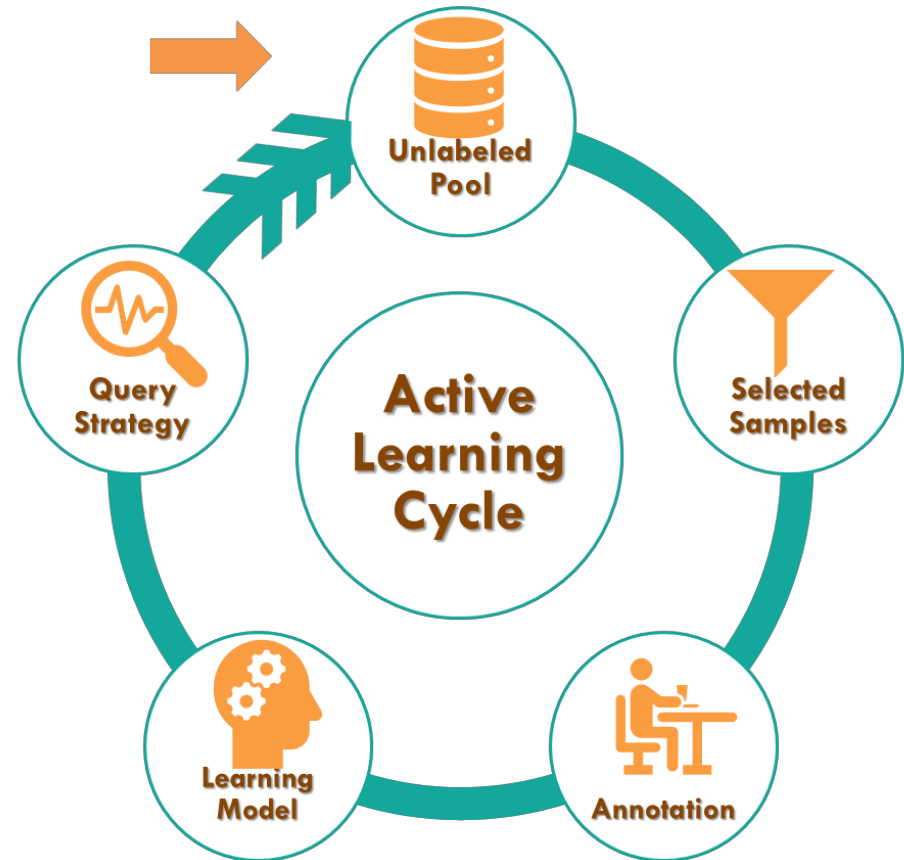
المساعدات الآلية

# Motivation

- ❖ Creating an annotated corpus is costly and time consuming.
- ❖ For example, developing *Curras*, a morphologically annotated corpus for the Palestinian dialect with approximately 56,000 words, took one year with the assistance of two full-time annotators.
- ❖ **Active learning** provides a cost-effective approach for developing annotated corpora with minimal annotation that can be used to develop multidialectal Arabic POS tagger.

# Active Learning

- ❖ Active learning means annotating while machine learning.
- ❖ Its work as:
  1. Select **initial sample** and annotate it.
  2. **Train** the model
  3. Select **new samples** by **query strategy** and annotate it.
  4. **Train** the model
  5. **Iterate** steps(3,4) until achieved **stopping point**.



# Our contribution

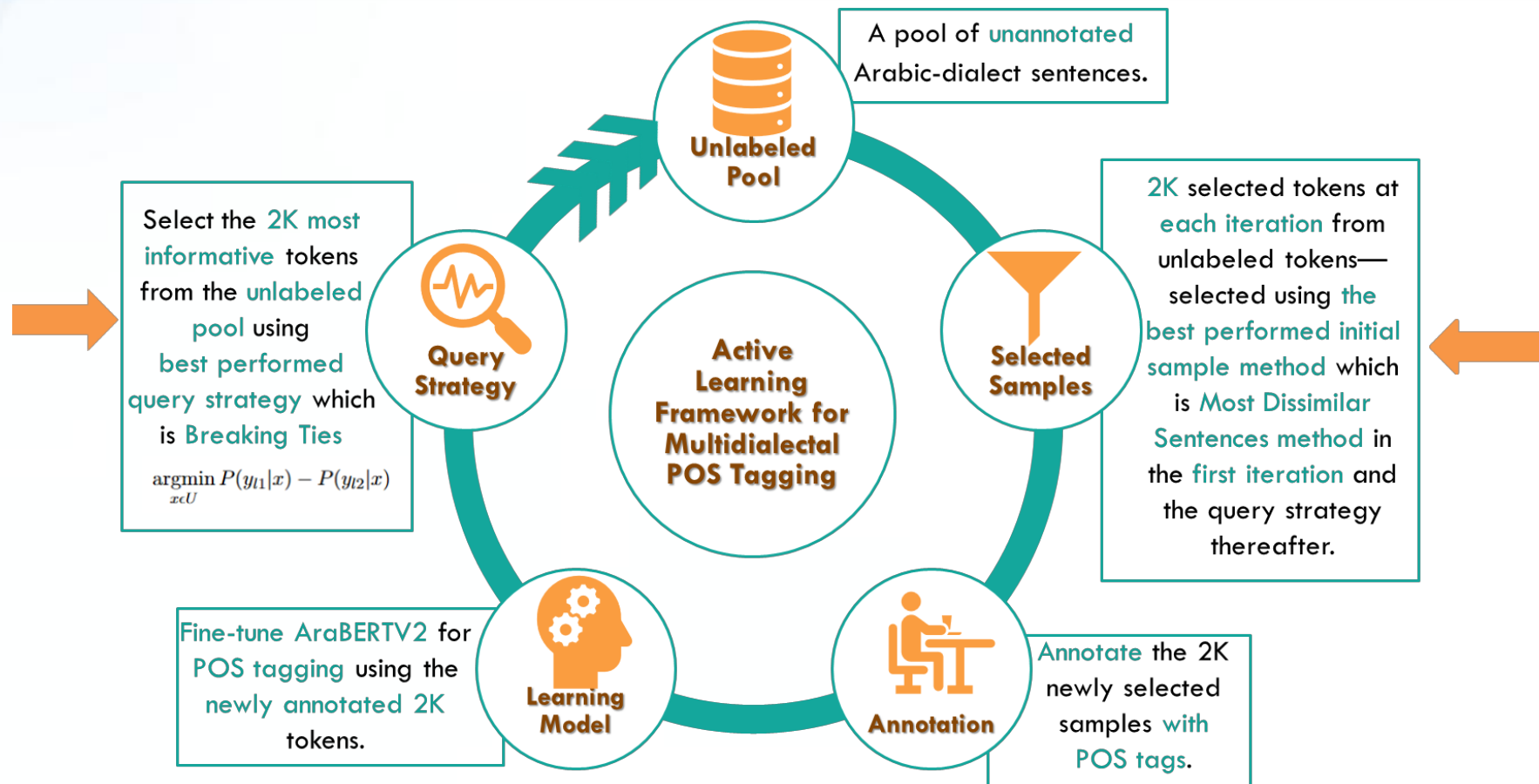
- ❖ Evaluation of active learning for multidialectal Arabic POS tagging (29 experiments).
- ❖ Insightful analysis and 10 assessments for initial sample selection across dialects.
- ❖ Active learning pipeline, e.g., achieved 97.6% with only 16,000 tokens in Emirati.

# Datasets Preparation and Composition

| Dataset                           | Tokens  | Unique Tokens |
|-----------------------------------|---------|---------------|
| ATB (MSA)                         | 339,710 | 51,820        |
| SALMA (MSA)                       | 34,253  | 8,718         |
| ARZTB (Egyptian)                  | 400,448 | 66,899        |
| Curras (Palestinian)              | 56,700  | 16,573        |
| Nabra (Syrian)+ Baladi (Lebanese) | 69,582  | 24,664        |
| Gumar (Emirati)                   | 201,596 | 22,924        |

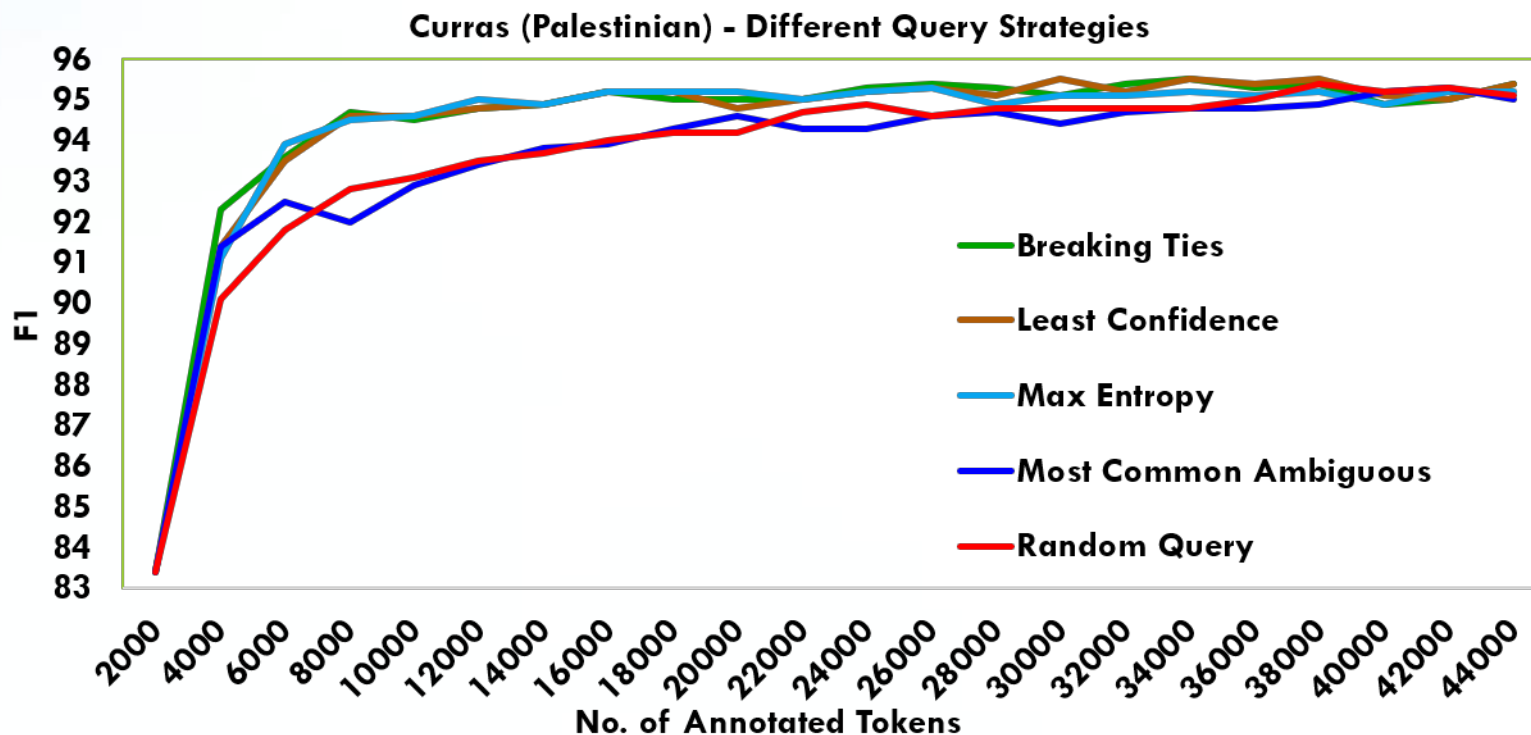
- ❖ **POS tags Mapping** : Label normalization, Value -level mapping
- ❖ **Splits to train, tune, dev, test**

# Active Learning Methodology

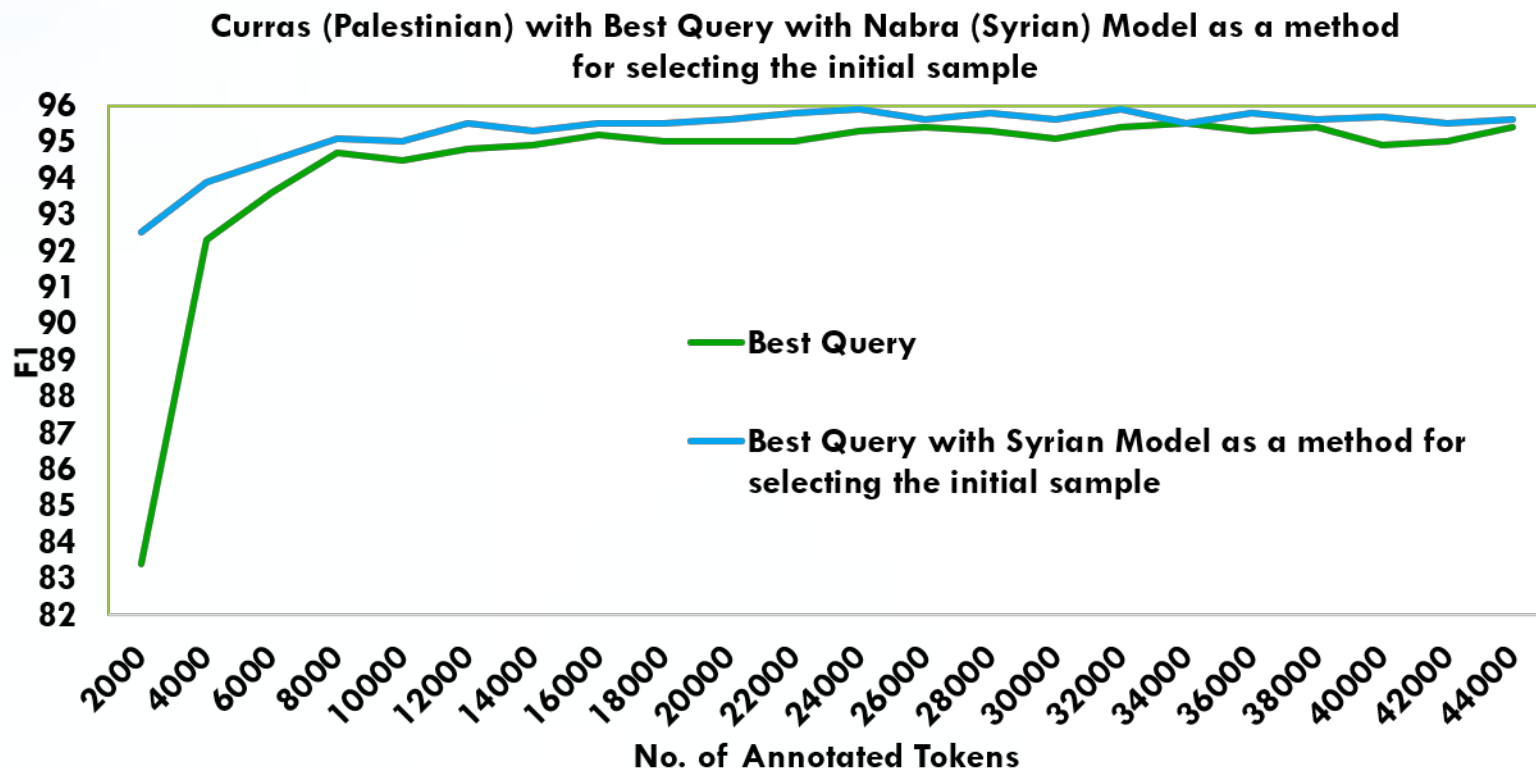


# Evaluation Results

# Query Strategy Evaluation

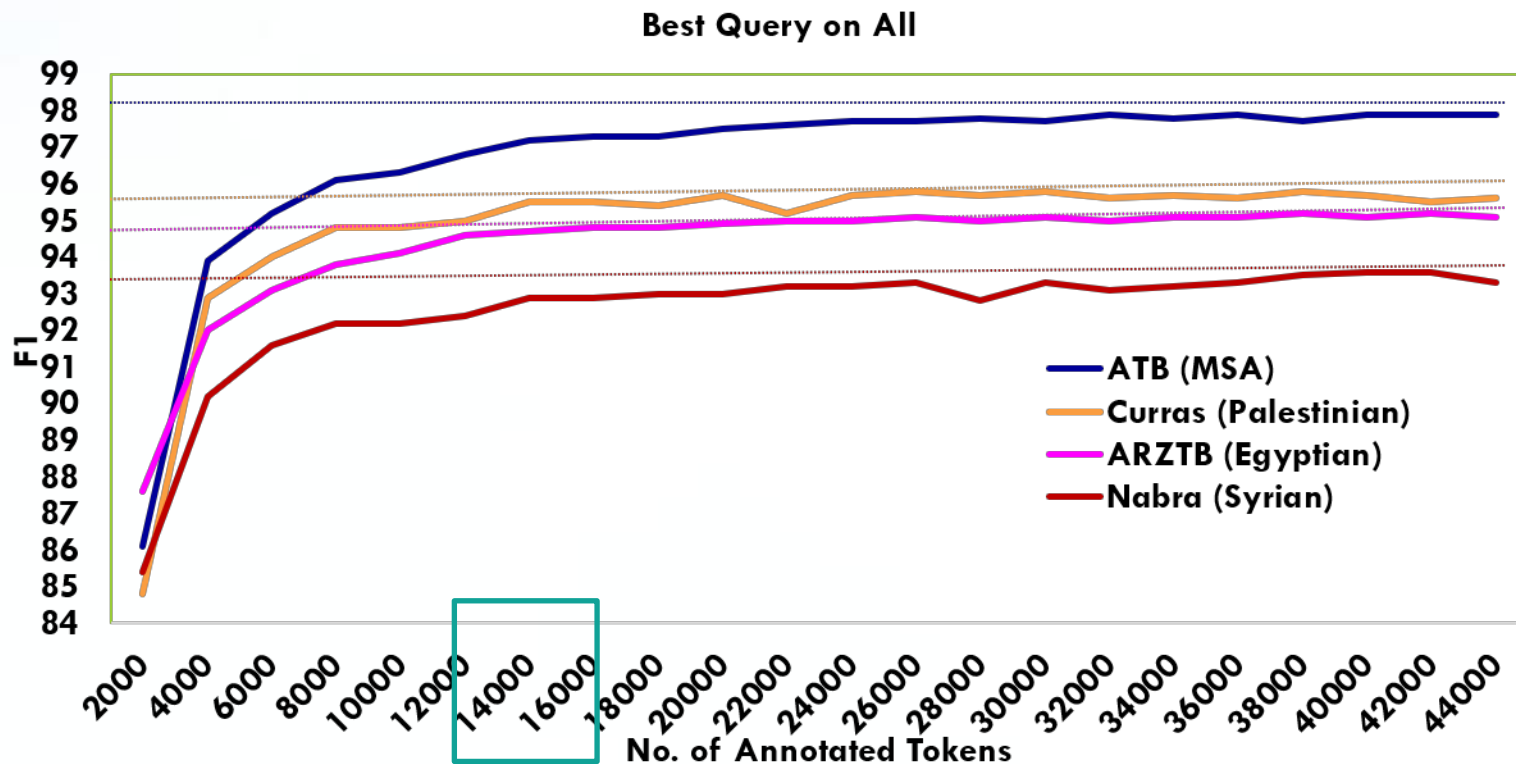


# Another-Dialect Model for Initial Sample Selection Evaluation



# Analysis Stabilization

# Analysis Stabilization



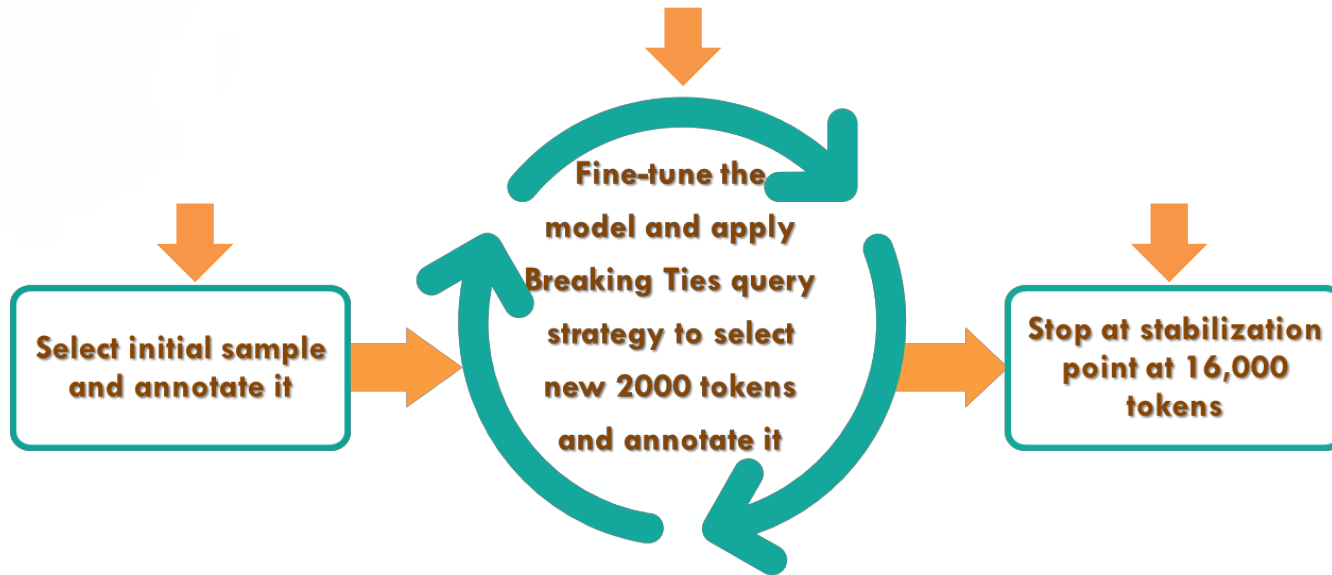
All Selected Tokens to Stabilize

- PART
- NOUN
- NOUN\_VOICE
- NOUN\_NUM
- NOUN\_PROP
- VERB
- NOUN\_QUANT
- REL\_PRON
- INTERROG\_PART
- FUT\_PART
- RESTRIC\_PART
- DET
- FOCUS\_PART
- SUB\_CONJ
- VERB\_PART
- PROG\_PART
- VOC\_PART
- NEG\_PART
- INTERJ
- ABBREV
- PREP
- PREP + SUB\_CONJ
- PSEUDO\_VERB
- ADJ
- ADJ\_NUM
- ADJ\_COMP
- PRON
- INTERROG\_PRON
- DEM\_PRON
- ADV
- DIGIT
- CONJ
- PUNC
- CV
- PV
- PV\_PASS
- IV
- IV\_PASS



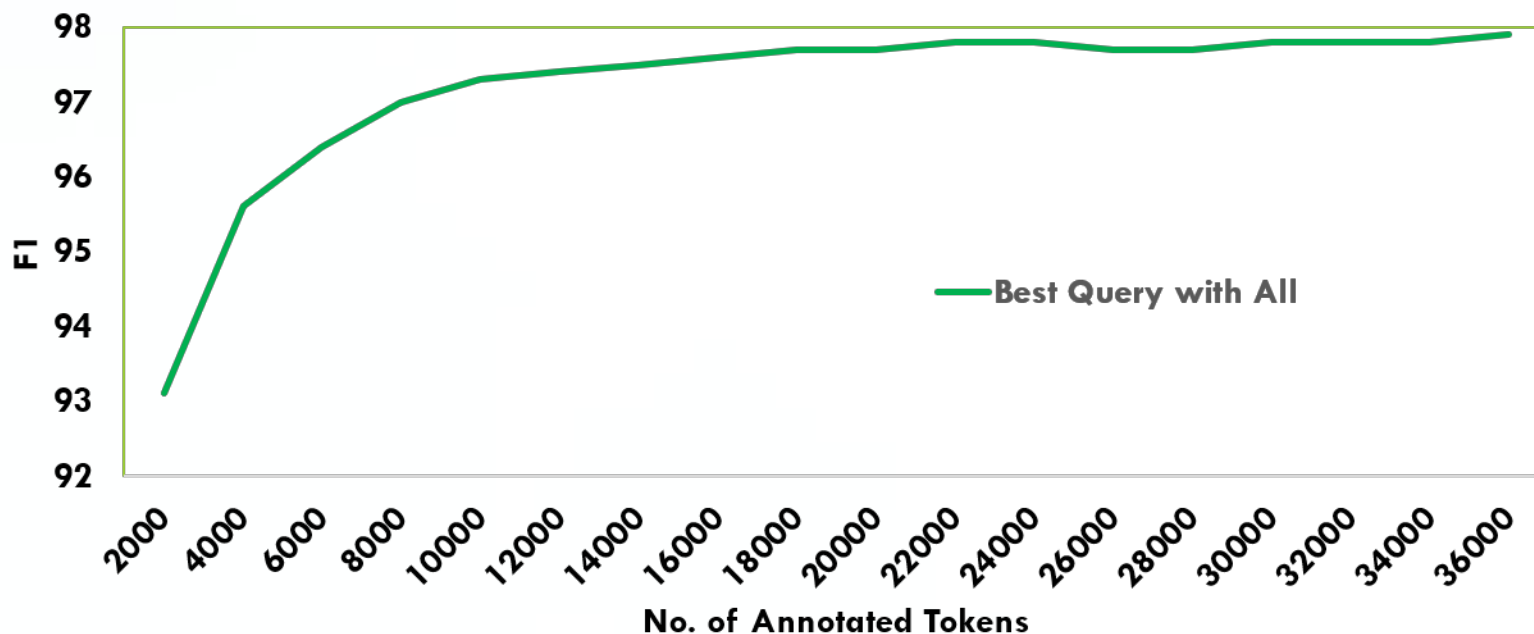
# Proposed Active Learning Pipeline

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# Proposed Active Learning Pipeline

**Gumar with Best Query with All (MSA+ Curras + Nabra + Egyptian)  
Model as a method for selecting the initial sample**



# Thank You

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